EXERCISE 7.5

(1) $\frac{1}{2}$ (2) 2 (3) ∞ (4) 1 (5) 0 (6) 0

(7) $\frac{-3}{2}$ (8) 1 (9) e (10) 1 (11) $\frac{1}{\sqrt{e}}$

EXERCISE 7.6

(1) (i) absolute maximum = -1, absolute minimum = -26

(ii) absolute maximum = 16, absolute minimum = -1

(iii) absolute maximum = 9, absolute minimum = $-\frac{9}{8}$

(iv) absolute maximum = $\frac{3\sqrt{3}}{2}$, absolute minimum = 0

(2) (i) strictly increasing on $(-\infty,-2)$ and $(1,\infty)$, strictly decreasing on $(-2,1)$

local maximum = 20 local minimum = -7

(ii) strictly decreasing on $(-\infty,5)$ and $(5,\infty)$. No local extremum.

(iii) strictly increasing on $(-\infty,\infty)$. No local extremum.

(iv) strictly decreasing on $(0,1)$, strictly increasing on $(1,\infty)$. local minimum = $\frac{1}{3}$

(v) strictly increasing on $\left(0,\frac{\pi}{4}\right),\left(\frac{3\pi}{4},\frac{5\pi}{4}\right)$, and $\left(\frac{7\pi}{4},2\pi\right)$.

strictly decreasing on $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$ and $\left(\frac{5\pi}{4},\frac{7\pi}{4}\right)$. local maximum= $\frac{11}{2}$ at $x=\frac{\pi}{4}$, $\frac{5\pi}{4}$.

local minimum= $\frac{9}{2}$ at $x=\frac{3\pi}{4}$, $\frac{7\pi}{4}$.

EXERCISE 7.7

(1) (i) concave upwards on $(-\infty,2)$ and $(4,\infty)$. Concave downwards on $(2,4)$

Points of inflection $(2,-16)$ and $(4,0)$

(ii) concave upwards on $\left(\frac{3\pi}{4},\frac{7\pi}{4}\right)$. Concave downwards on $\left(0,\frac{3\pi}{4}\right)$ and $\left(\frac{7\pi}{4},2\pi\right)$

Points of inflection $\left(\frac{3\pi}{4},0\right)$ and $\left(\frac{7\pi}{4},0\right)$

(iii) concave upwards on $(0,\infty)$. Concave downward on $(-\infty,0)$

Points of inflection $(0,0)$

(2) (i)local minimum = -2 ; local maximum = 2 (ii)local minimum = $-\frac{1}{e}$

(iii)local minimum = 0 ; local maximum = $\frac{1}{e^2}$

(3) strictly increasing on $(-\infty,-1)$ and $\left(\frac{1}{2},\infty\right)$. strictly increasing on $\left(-1,\frac{1}{2}\right)$

local maximum = 6 , local minimum = $-\frac{3}{4}$

concave downwards on $\left(-\infty,-\frac{1}{4}\right)$; concave upwards on $\left(-\frac{1}{4},\infty\right)$.

point of inflection $\left(-\frac{1}{4},\frac{21}{8}\right)$

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